

Introduction to Multimedia Security and Privacy

Course Overview and Structure

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Copyright Notice

This course uses in part some materials from:

- **Book:** I. Cox, M. Miller, and J. Bloom, "*Digital Watermarking: Principles and Practice*", Morgan Kaufmann Publisher Inc., San Francisco, 2001.
- **Book:** M. Barni and F. Bartolini, "*Watermarking Systems Engineering: Enabling Digital Assets Security and Other Applications*", Marcel Dekker, 2004.
- **Book:** J. Fridrich, "*Steganography in Digital Media: Principles, Algorithms, and Applications*", Cambridge University Press, 2009.

Additional Sources:

- Papers, books, and presentations that will be indicated along the course.

Usage Restrictions:

The usage of these slides for commercial or educational purposes is prohibited without the authorization of the course leaders and the permission of the above document authors.

Grading Policy

- The final grade for this course will be determined based on:

Grading Breakdown

- **Homework Assignments:** not evaluated
- **Final Examination:** oral exam
 - The final exam will be an **oral examination** conducted at the end of the course.

Important Note

It is **strongly advised** to complete all homework assignments to build a strong foundation and secure corresponding grades, as they contribute significantly to the final evaluation.

Python Environment for the Class

- Install Python (preferably version 3.9 or higher).
- Recommended tools:
 - IDE: PyCharm, Jupyter Notebook, or Google Colab.
 - Packages:
 - `numpy`, `opencv-python`, `matplotlib`
 - `scikit-image`, `scipy`
- Ensure you can execute Python scripts and interpret code.
- Homework and repositories:
 - Repositories will provide:
 - Slides
 - Sample code and datasets
 - Recorded video lectures

Course Structure

The course is divided into four main blocks:

- ❶ **Block 1: Digital Watermarking**
- ❷ **Block 2: Perceptual Hashing**
- ❸ **Block 3: Security of Foundation Models**
- ❹ **Block 4: Privacy Preservation**
- ❺ **Block 5: Physical World Security** — optional as a bonus

Learning Objectives

- Understand foundational techniques in multimedia security.
- Analyze modern challenges and applications.
- Apply robust solutions to real-world problems.

Block 1: Digital Watermarking

Definition

Digital Watermarking embeds imperceptible information into multimedia content (e.g., images, audio, video) to protect copyright, authenticate content, or enable secure communication.

- Originates from **steganography**—the art of secret writing.
- Embedding is typically based on a **secret key**.
- Ensures:
 - **Ownership verification**
 - **Robustness to attacks**
 - **Content integrity**

Block 1: Block diagram of Digital Watermarking

Visible



Hidden



Block 1: Applications of Digital Watermarking

- **Robust Watermarking:**

- Embedding imperceptible signals resilient to distortions.
- Applications: Copyright protection, distinguishing real from synthetic data.

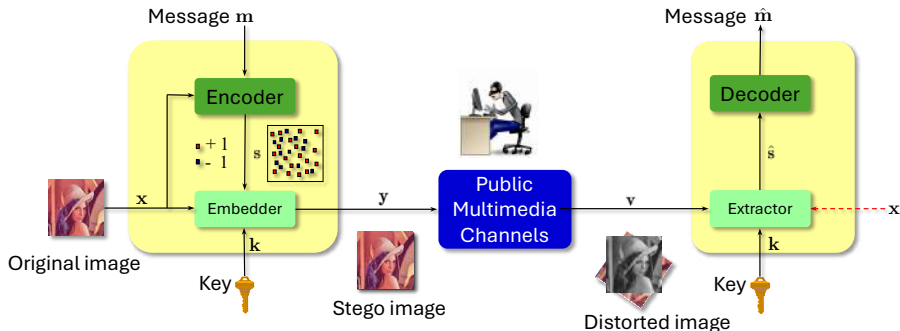
- **Steganography:**

- Hiding secret messages in multimedia content.
- Applications: Secure communication without revealing the existence of hidden information.

- **Tamper Detection:**

- Verifying integrity of content by detecting modifications.
- Applications: Digital forensics, image authentication.

Block 1: Block diagram of Digital Watermarking

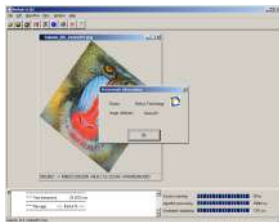


Attacks on:

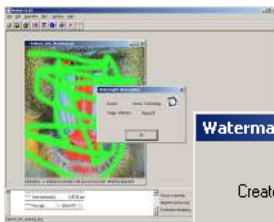
- **Robustness:** Damage message applying distortions
- **Security:** Damage message or guess key applying adversarial attacks

Block 1: Robust Digital Watermarking

Robust watermarking



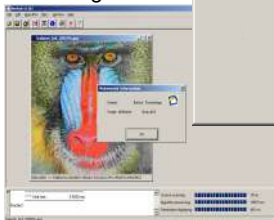
Rotation/shearing



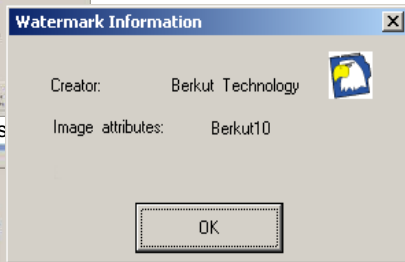
Drawing/erasures



Scaling/aspect ratio



Lossy compression JPEG/JPEG2K

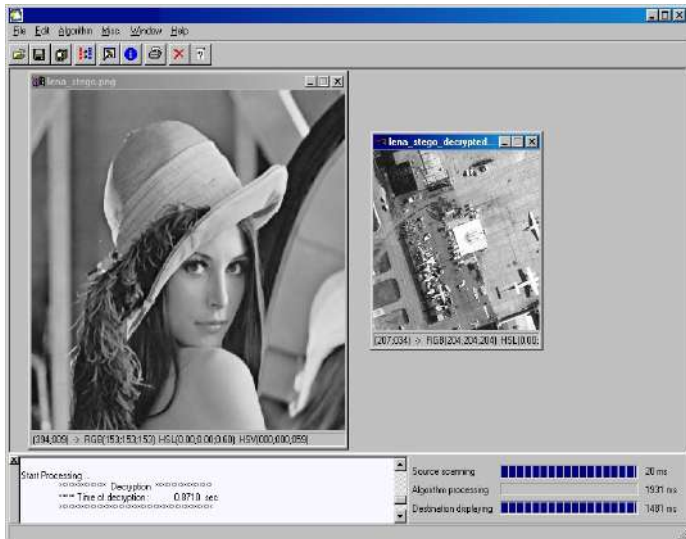


Block 1: Robust Digital Watermarking

- **Definition:** Ensures the identification of the **owner** and tracking of content.
- **Requirements:**
 - Robustness: Must withstand JPEG compression, rotation, cropping, and adversarial attacks.
 - Perceptibility: Should remain invisible to retain commercial value, especially in the era of generative AI.
 - Embedding Capacity: Typically 64–128 bits per image across three channels.
- **Applications:**
 - Copyright protection.
 - Differentiating synthetic from real content.
- **Key Management:** Symmetric keys required for detection or decoding.
- **Access to Original Content:**
 - Generally not required, but some AI-based or protocol-specific solutions may need it.
- **Compliance:** Must adhere to EU regulations.

Block 1: Steganography

Steganography

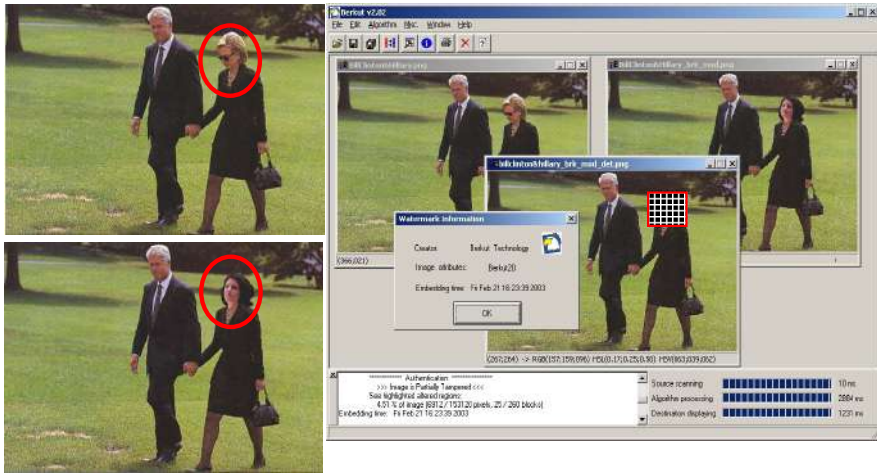


Block 1: Steganography

- **Definition:** Enables **secret communication** by hiding the existence of communication itself.
- **Requirements:**
 - Robustness: Not critical; adversarial robustness is relaxed.
 - Statistical Undetectability: Crucial for concealing the fact of communication.
 - Embedding Capacity: Allows kilobytes per image; information can be distributed across multiple images.
- **Applications:**
 - Alternative to cryptography when its use is undesirable or prohibited.
- **Key Management:** Steganographic keys are required for secure embedding and detection.
- **Statistical Detectability:** Focus on avoiding detection by machine learning-based steganalysis.

Block 1: Tamper Detection

Tamper proofing



Block 1: Tamper Detection

- **Definition:** Detects and localizes **perceptual modifications** in images.
- **Requirements:**
 - Key: Ensures regions correspond to the original content.
 - Capacity: Relatively high to embed sufficient information.
 - Robustness: A trade-off with capacity, as embedding more information reduces robustness.
- **Applications:**
 - Content verification.
 - Forensic analysis.
- **Techniques:**
 - Key-based: Embedding with a symmetric key for regional checks.
 - Compression-based: Hashing or compressing original data and embedding it.

Block 1: Challenges and Trade-Offs in Digital Watermarking

- Trade-offs include:
 - **Robustness**: Resilience to compression, noise, and geometric transformations (e.g., rotations, cropping, shearing).
 - **Capacity**: Embedding sufficient information while maintaining quality.
 - **Perceptual Invisibility**: Ensuring the watermark is imperceptible to humans.
 - **Security**: Preventing adversarial and leakage attacks.
- Techniques include:
 - **Zero-bit watermarking**: Detection problem (yes/no).
 - **M-bit watermarking**: Decoding problem (extracting bits).
 - Implementation domains:
 - Additive and quantization methods.
 - Geometrical robustness via self-synchronization and foundation models.

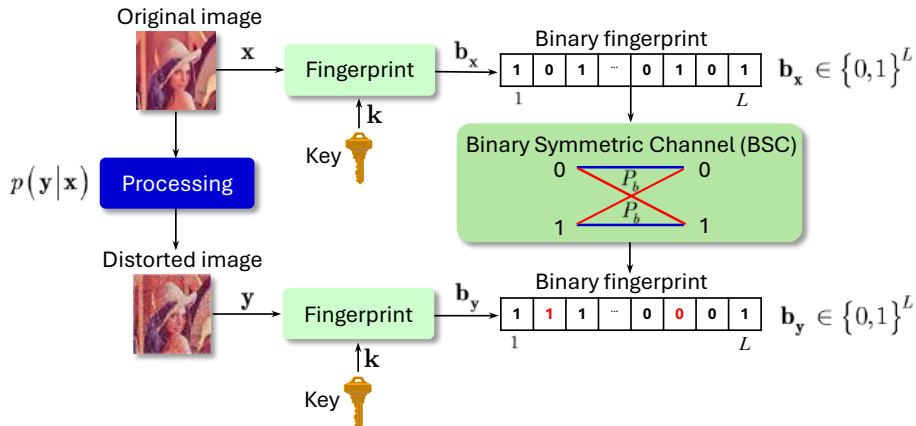
Block 2: Perceptual Hashing

Definition

Perceptual Hashing (aka robust hashing and content fingerprinting) generates a compact representation of multimedia content that is robust to distortions while remaining similar to the original hash.

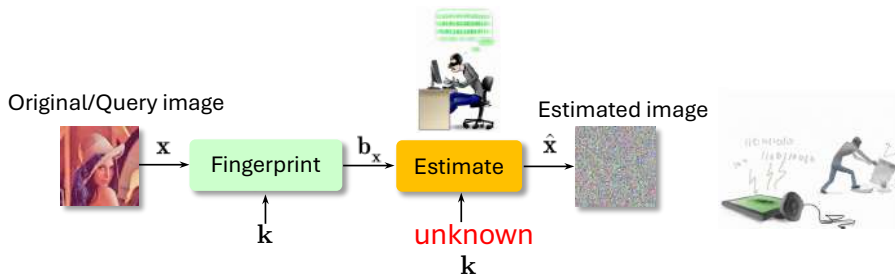
- Applications:
 - **Privacy-Preserving Retrieval:** Securely searching for similar multimedia content.
 - **Content Authentication:** Identifying tampered content.
 - **Vector Databases:** Efficient storage of neural network features for large-scale retrieval.

Block 1: Block diagram of Perceptual Hashing



Block 1: Irreversibility of Perceptual Hashing

Irreversibility



Block 2: Perceptual Hashing

- **Robustness:**

- Must handle **image processing** challenges such as compression, additive noise variation, and illumination conditions.
- Must also be robust to **geometric transformations** like rotation, shearing, and scaling.

- **Compactness:**

- Hashes should be **short and compact**, often using **quantized representations** such as vector quantization and binary quantization.

- **Security:**

- Should resist **adversarial attacks** and hash forgery: content may be modified but produce the same hash (**false negatives**), or different content may lead to identical hashes (**false positives**).

- **Irreversibility:**

- The hash should **not reveal** any information about the original content, which is crucial for privacy-preserving biometrics, multimedia retrieval, and secure storage.

- **Compliance:**

- Must adhere to **regulations** for secure and ethical usage (Responsible AI).

Why Are Traditional Security Tools Not Suitable for Multimedia?

- **Traditional Security:**

- **Cryptographic Encryption:** Ensures confidentiality of information.
- **Cryptographic Hashes:** Used for authentication, trust, and access control.

- **Challenges with Classical Cryptography:**

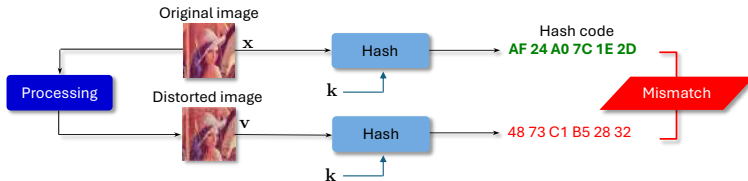
- Highly sensitive to noise and unintentional distortions in input data.
- Designed for binary, non-perceptual data (zeros and ones).
- Not robust to transformations or cross-domain conversions (e.g., images, audio).

- **Why We Need Digital Watermarking and Perceptual Hashing:**

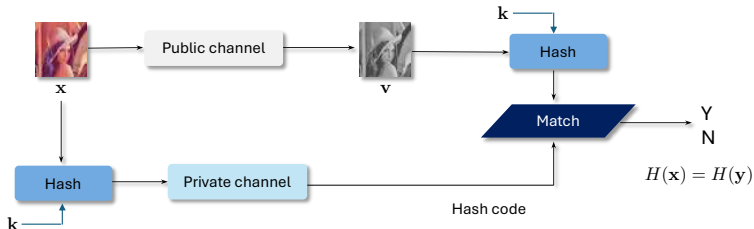
- Multimedia data is inherently noisy and undergoes distortions (e.g., compression, cropping, rotation).
- Perceptual similarity, rather than exact matches, is crucial for security and privacy.
- Applications extend to biometrics, copyright protection, and secure communications.

Why Are Traditional Security Tools Not Suitable for Multimedia?

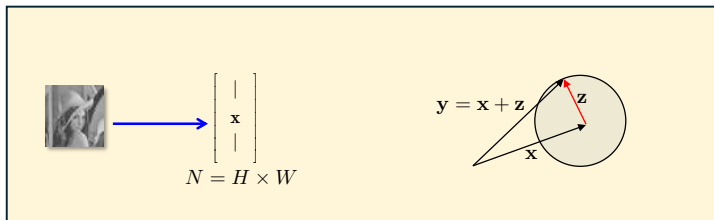
Main problem



Practical implementation



Why Are Traditional Security Tools Not Suitable for Multimedia?



Crypto hashes

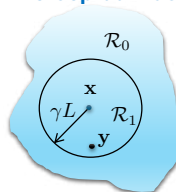
$$\mathbf{x} = \mathbf{y}$$


Decision region

$$H(\mathbf{x}) = H(\mathbf{y})$$

$$\|\mathbf{y} - \mathbf{x}\|_2^2 = 0$$

Perceptual hashes



Decision region

$$\|\mathbf{y} - \mathbf{x}\|_2^2 \leq \gamma L$$

Challenges with Traditional Privacy Protection

- **Common Methods for Privacy Protection:**

- **Data Degradation and Randomization Techniques:**

- Noise addition.
 - Lossy compression.
 - Dimensionality reduction.
 - Data removal.

- **Anonymization Techniques:**

- Differential privacy.
 - Randomized rules.

- **Why Traditional Privacy Methods Are Insufficient?**

- Modern AI techniques, such as generative models, can regenerate degraded or anonymized data.
 - Example: Reconstruction of biometric templates, such as faces or fingerprints, using AI generative models.
 - Partial information leakage can compromise privacy significantly.

- **Need for Robust Privacy Solutions:**

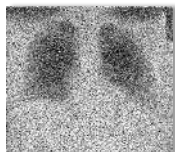
- Techniques that ensure security even under partial disclosure or attack.
 - Use of robust perceptual hashing, advanced cryptographic techniques, and AI-driven protections.

Why Traditional Privacy Methods Are Insufficient?

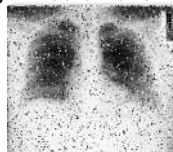
Original



50% removed based on key



Median 9x9



Adaptive recovery



Block 3: Security of Foundation Models

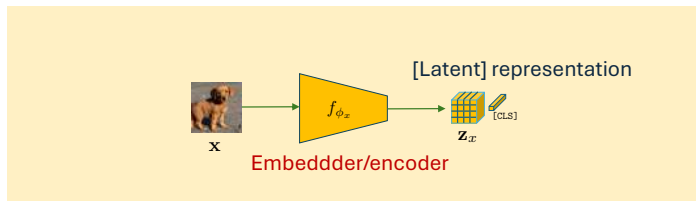
Definition

Foundation Models are large-scale neural networks trained on diverse datasets to learn robust representations for tasks like image classification, text generation, and audio analysis.

- Types of foundation models:
 - Representation learning models.
 - Generative models for text, images, and video.
- Challenges:
 - Vulnerability to adversarial perturbations.
 - Potential misuse (e.g., deepfakes, illegal image modification).

Block 3: Security of Foundation Models

Main idea of representation learning



Block 3: Security of Foundation Models

Definition

Security of foundation models is defined as their **robustness to adversarial attacks**.

What is an Adversarial Attack?

An adversarial attack targets deep learning systems by introducing small, imperceptible modifications to the input, exploiting the system's sensitivity and differentiability to cause drastic variations in the output.

Key Characteristics of Adversarial Attacks:

- Specific attacks targeting the impairment of normal model performance.
- Input modifications are typically invisible to human perception.
- Aim to cause dramatic performance degradation in the foundation models.

Importance of Security in Foundation Models

Why is Security Important?

- Foundation models are widely used in critical applications such as:
 - Digital watermarking.
 - Biometrics.
 - Multimedia security.
 - Retrieval systems.

Two Key Adversarial Attack Cases:

Case 1: Perceptually Similar Images

Perceptually similar images (e.g., original $\mathbf{x}_0$ and attacked $\mathbf{x}_A$) have very different perceptual hashes, resulting in **low cosine similarity** in the latent space.

Case 2: Perceptually Dissimilar Images

Perceptually dissimilar images (e.g., original $\mathbf{x}_0$ and attacked $\mathbf{x}_A$) have **high cosine similarity** in the latent space, leading to false matches or erroneous recognition.

Adversarial Attack Case 1: Perceptually Similar Inputs

Case 1: Adversarially attacked images are **perceptually similar** but lead to **different perceptual hashes**.

$\mathbf{x}_0$: Original Image, $\mathbf{x}_A$: Attacked Image

Characteristics:

- $\mathbf{x}_0$ and $\mathbf{x}_A$ are close in Mean Squared Error (MSE).
- However, their perceptual hashes $\mathbf{z}_{\mathbf{x}_0}$ and $\mathbf{z}_{\mathbf{x}_A}$ have **low cosine similarity**.
- This causes significant divergence in retrieval, recognition, or other tasks relying on perceptual hashing.

Implications:

- The attack compromises the robustness of the model to small, imperceptible perturbations.

Adversarial Attack Case 2: Perceptually Dissimilar Inputs

Case 2: Adversarially attacked images are **perceptually dissimilar** but lead to **similar perceptual hashes**.

$\mathbf{x}_0$: Original Image, $\mathbf{x}_A$: Attacked Image

Characteristics:

- $\mathbf{x}_0$ and $\mathbf{x}_A$ are significantly different perceptually.
- However, their perceptual hashes $\mathbf{z}_{\mathbf{x}_0}$ and $\mathbf{z}_{\mathbf{x}_A}$ have **high cosine similarity**.
- This causes false matches or erroneous retrieval results.

Implications:

- The attack exploits vulnerabilities in the latent space representation of the foundation model.
- Critical for tasks requiring secure content representation (e.g., biometrics, security).

Block 4: Privacy Preservation

Definition

Privacy Preservation involves securing sensitive data such as biometric identifiers against misuse while maintaining usability for authentication or retrieval.

- Techniques:
 - Helper database methods for secure retrieval.
 - Group testing for privacy protection.
 - Fuzzy commitment schemes for robust biometric storage.
- Applications:
 - Border control
 - Financial transactions

Conclusion

- This course provides a comprehensive foundation in multimedia security and privacy.
- Key focus:
 - Robustness and security of multimedia techniques.
 - Practical applications in watermarking, perceptual hashing, and privacy preservation.
- **Structure of the course:**
 - Introduction to theory and challenges.
 - Detailed mathematical formulations.
 - Python-based examples for implementation and visualization.
- **Outcome:**
 - Gain theoretical and practical skills in multimedia security.